

A spectrum of a triangulated category with an action

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Introduction

Motivation

All schemes are assumed to be noetherian.

X : scheme, $\alpha \in \mathrm{dgBr}(X)$,

$\mathrm{Perf}(X, \alpha)$: twisted derived category of perfect complexes on X .

There is a natural action of $\mathrm{Perf}(X)$ on $\mathrm{Perf}(X, \alpha)$.

Example 1

$\{\text{thick subcategory of } \mathrm{Perf}(X, \alpha) \text{ closed under the action of } \mathrm{Perf}(X)\}$
 $\longleftrightarrow \{\text{specialization-closed subsets of } X\}.$

Motivation 2

X : scheme,

L : line bundle, $W \in \Gamma(X, L)$,

X_0 : zero scheme of W ,

$\mathrm{DMF}(X, L, W)$: derived matrix factorization category.

There is a natural action of $\mathrm{DMF}(X, L, 0)$ on $\mathrm{DMF}(X, L, W)$.

Assume that X is separated, admits an ample family of line bundles, and that W is not a zero-divisor.

Example 2

Hirano (2019)

$\{\text{thick subcategory of } \mathrm{DMF}(X, L, W) \text{ closed under the action of } \mathrm{DMF}(X, L, 0)\}$
 $\longleftrightarrow \{\text{specialization-closed subsets of } \mathrm{Sing}(X_0/X)\},$

where $\mathrm{Sing}(X_0/X)$ is the relative singular locus.

Motivation 3

X : locally complete intersection scheme,

$\mathrm{Coh}(X)$: bounded derived category of coherent sheaves on X ,

$\mathrm{Sing}(X) = \mathrm{Coh}(X)/\mathrm{Perf}(X)$,

X^{coh} : certain topological space,

$X^{\mathrm{sing}} = X^{\mathrm{coh}} \setminus X$.

Example 3

Stevenson (2014), Liu–Pollitz (2025)

$\{\text{thick subcategory of } \mathrm{Coh}(X) \text{ closed under the action of } \mathrm{Perf}(X)\}$

$\longleftrightarrow \{\text{specialization-closed subsets of } X^{\mathrm{coh}}\},$

$\{\text{thick subcategory of } \mathrm{Sing}(X) \text{ closed under the action of } \mathrm{Perf}(X)\}$

$\longleftrightarrow \{\text{specialization-closed subsets of } X^{\mathrm{sing}}\}.$

Motivation 4

What we want to do:

A triangulated category with an action from a tt-category

$\longmapsto$ topological space which classifies thick subcategories closed under the action.

Main results

- ① Constructed $\mathrm{Spec}(T, S)$!
 - ① A relative version of Matsui's triangular spectrum.
 - ② Defined for triangulated categories with an action of a tt-category.
 - ③ Classifies thick subcategories closed under the action.
 - ④ Larger base $\rightsquigarrow$ smaller spectrum, and smaller base $\rightsquigarrow$ larger spectrum.
 - ⑤ For some categories, it proves an analog of tt-spectra.
- ② Universality.
- ③ Comparisons with other spectra.
 - ① Generalizes both Balmer's and Matsui's spectra.
- ④ Some properties.
 - ① Morphism to the spectrum of the base tt-category.
 - ② The set of submodules becomes a sheaf on the base space.
 - ③ Fiberwise decomposition.

Twisted derived category of perfect complexes

X : scheme, $\alpha \in \mathrm{dgBr}(X)$.

Example 4

There is a canonical isomorphism

$$\mathrm{Spec}(\mathrm{Perf}(X, \alpha), \mathrm{Perf}(X)) \cong X$$

and a bijection

$$\mathrm{Sub}_{\mathrm{Perf}(X)}(\mathrm{Perf}(X, \alpha)) \longleftrightarrow \{\text{specialization-closed subsets of } X\}.$$

Derived matrix factorization category

X : scheme,

L : line bundle, $W \in \Gamma(X, L)$,

X_0 : zero scheme of W .

Example 5

Assume that X is separated, admits an ample family of line bundles, and that W is not a zero-divisor. Then there is a homeomorphism

$$\mathrm{Spec}^{\mathrm{fin}}(\mathrm{DMF}(X, L, W), \mathrm{DMF}(X, L, 0)) \cong \mathrm{Sing}(X_0/X)$$

and a bijection

$$\mathrm{Sub}_{\mathrm{DMF}(X, L, 0)}(\mathrm{DMF}(X, L, W)) \longleftrightarrow \{\text{specialization-closed subsets of } \mathrm{Sing}(X_0/X)\}.$$

The categories related to singularities

X : locally complete intersection scheme,

Example 6

$$\mathrm{Spec}(\mathrm{Sing}(X), \mathrm{Perf}(X)) \cong X^{\mathrm{sing}},$$

$$\mathrm{Spec}(\mathrm{Coh}(X), \mathrm{Perf}(X)) \cong X^{\mathrm{coh}},$$

$$\mathrm{Sub}_{\mathrm{Perf}(X)}(\mathrm{Coh}(X)) \longleftrightarrow \{\text{specialization-closed subsets of } X^{\mathrm{coh}}\},$$

$$\mathrm{Sub}_{\mathrm{Perf}(X)}(\mathrm{Sing}(X)) \longleftrightarrow \{\text{specialization-closed subsets of } X^{\mathrm{sing}}\}.$$

Spectrum of triangulated categories

	tt-sp.	relative $\triangle$ -sp.	$\triangle$ -sp.
Defined for	tt-cat.	$\triangle$ -cat. with action	$\triangle$ -cat.
Size	small	depends on the base	large
Computability	relatively easy	depends on the base	hard
Functoriality	functorial	morphism to the base	none
Classifies	$\otimes$ -ideals	submodules	thick subcategories
$\text{Perf}(X)$	X	relative FM partners	FM partners

Construction and universality

Triangulated category with action

T : triangulated category

S : tensor triangulated category

$- \otimes - : S \times T \rightarrow T$

Definition 7

T is an S -module if:

- 1 For any $t \in T$, the functor $- \otimes t : S \rightarrow T$ is triangulated.
- 2 For any $s \in S$, the functor $s \otimes - : T \rightarrow T$ is triangulated.
- 3 For $1_S \in S$, the functor $1_S \otimes - : T \rightarrow T$ is naturally equivalent to the identity.
- 4 There is a natural isomorphism $(s_1 \otimes s_2) \otimes t \cong s_1 \otimes (s_2 \otimes t)$ for $s_1, s_2 \in S$, $t \in T$.

Submodules and prime submodules

T : S -module

Definition 8

- ① A subcategory $I \subset T$ is a S -submodule if it is thick and closed under the S -action.
 $\text{Sub}_S(T) := \{S\text{-submodules}\}.$
- ② A S -submodule $P \subset T$ is prime if $\{J \in \text{Sub}_S(T) \mid J \supsetneq P\}$ has the smallest element.
 $\text{Spec}(T, S) := \{S\text{-prime submodules}\}.$

Support and topology

For a subset $W \subset T$, define:

$$\mathrm{Supp}(W) := \{P \in \mathrm{Spec}(T, S) \mid W \not\subset P\}.$$

We endow $\mathrm{Spec}(T, S)$ with the weakest topology such that $\mathrm{Supp}(a)$ is closed for all $a \in T$.

Definition 9

The topological space $\mathrm{Spec}(T, S)$ is called the relative triangular spectrum of the S -module T .

Classification

T : S -module, $I \in \text{Sub}_S(T)$.

Lemma 10

Let $F \in T \setminus I$. Then there exists a prime S -submodule P such that $I \subset P$ and $F \notin P$.

It follows from Zorn's lemma.

Classification

S -submodules are classified by their supports.

Theorem 11

The map

$$\mathrm{Sub}_S(T) \rightarrow 2^{\mathrm{Spec}(T,S)}, \quad I \mapsto \mathrm{Supp}(I)$$

is injective. The left inverse is

$$Z \mapsto \{a \in T \mid \mathrm{Supp}(a) \subset Z\}.$$

The same holds for Matsui's triangular spectrum and thick categories.

Example: Severi-Brauer scheme of relative dimension 1

$\text{Supp} : \text{Sub}_S(T) \rightarrow \{\text{specialization-closed subsets of } \text{Spec}(T, S)\}$

is not surjective in general.

Y : scheme X : Severi-Brauer scheme over Y of relative dimension 1

Example 12

$$\text{Spec}(\text{Perf}(X), \text{Perf}(Y)) \cong X \amalg \left(\coprod_{\mathbb{Z}} Y \right).$$

X on the right corresponds to the canonical open immersion. Each copy of Y corresponds to the admissible subcategory $\text{Perf}(X, Y)_n \subset \text{Perf}(X)$, which is equivalent to $\text{Perf}(Y, \alpha^n)$ for some $\alpha \in \text{dgBr}(Y)$.

Example: Severi-Brauer scheme of relative dimension 1

$\text{Sub}_{\text{Perf}(Y)}(\text{Perf}(X))$ corresponds bijectively to specialization-closed subsets $Z \subset X \amalg (\amalg_{\mathbb{Z}} Y)$ with the following fiberwise description: for every point in Y , the fiber Z_y satisfies exactly one of the followings:

- 1 $Z_y = \emptyset$.
- 2 $Z_y = W \amalg (\amalg_{\mathbb{Z}} \{y\})$ for some non-empty specialization-closed $W \subset X_y$.
- 3 $Z_y = X_y \amalg (\amalg_{\mathbb{Z} \setminus \{n\}} \{y\})$ for some $n \in \mathbb{Z}$.

$\text{Sub}_{\text{Perf}(Y)}(\text{Perf}(X)) \rightarrow \{\text{specialization-closed subsets of } \text{Spec}(\text{Perf}(X), \text{Perf}(Y))\}$

is not surjective.

Support datum

Definition 13

A support datum of a S -module T is a T_0 -topological space X with a map $\text{Supp} : \text{Obj}(T) \rightarrow 2^X$ satisfying:

- ① (shift invariance) $\text{Supp}(a) = \text{Supp}(a[1])$.
- ② (action) $\text{Supp}(s \otimes a) \subset \text{Supp}(a)$.
- ③ (direct sums) $\text{Supp}(a \oplus b) = \text{Supp}(a) \cup \text{Supp}(b)$.
- ④ (exactness) If $a \rightarrow b \rightarrow c$ is exact, then $\text{Supp}(b) \subset \text{Supp}(a) \cup \text{Supp}(c)$.
- ⑤ (distinguishability) $\bigcup_{a \in I} \text{Supp}(a) \neq \bigcup_{b \in J} \text{Supp}(b)$ for $I \neq J \in \text{Sub}_S(T)$.
- ⑥ (topology) The topology is generated by $\{X \setminus \text{Supp}(a) \mid a \in T\}$.

Universality

Theorem 14

$(\mathrm{Spec}(T, S), \mathrm{Supp})$ is the universal support datum: for any support datum (X, Supp_X) , there exists a unique continuous map $f : \mathrm{Spec}(T, S) \rightarrow X$ such that $f^{-1}(\mathrm{Supp}_X(a)) = \mathrm{Supp}(a)$ for all $a \in T$.

A similar statement holds for Matsui's triangular spectrum with the action condition removed and the other conditions unchanged.

Generalization of support datum

A semi-lattice L is a partially ordered set where any two elements have a supremum.

A subset $S \subset L$ is an ind-object if:

- 1 $S \neq \emptyset$.
- 2 If $I \leq s$ with $s \in S$, then $I \in S$.
- 3 $\text{Sup}\{s, s'\} \in S$ for all $s, s' \in S$.

$$\text{Ind}(L) := \{S \subset L \mid S \text{ is an ind-object}\}.$$

Generalization of support datum 2

Definition 15

A support datum of L is a T_0 -space X with a semi-lattice map $\text{Supp} : L \rightarrow 2^X$ satisfying:

- ① (distinguishability) The induced map $\text{Supp} : \text{Ind}(L) \rightarrow 2^X$ is injective.
- ② (topology) The closed subsets are generated by $\{\text{Supp}(I) \mid I \in L\}$.

Generalization of support datum 3

$$X_L = \{P \in \text{Ind}(L) \mid \bigcap_{S \supsetneq P} S \not\supsetneq P\}, \quad \text{Supp}(I) = \{P \in X_L \mid I \notin P\}.$$

Theorem 16

For any semi-lattice L , (X_L, Supp) is the universal support datum.

(proof)

Similarly to the proof of the previous theorem, one can check that (X_L, Supp) is a support datum.

For any support datum (X, Supp) , there is an injective map

$$\iota : X \rightarrow 2^L, \quad x \mapsto \{I \in L \mid x \notin \text{Supp}(I)\}.$$

The universal property is proved by using this map.

Example

S : set,

$L = \{S' \subset S \mid \text{finite}\}.$

Example 17

$$\text{Ind}(L) = S$$

$$X_L = \{S' \subset S \mid \#(S \setminus S') = 1\} \cong S$$

Connection with relative spectrum

Let T be an S -module and $L = \{\text{finitely generated } S\text{-submodules}\}$. Then,

$$\begin{aligned}\mathrm{Ind}(L) &\cong \mathrm{Sub}_S(T), \\ X_L &\cong \mathrm{Spec}(T, S).\end{aligned}$$

Moreover, isomorphism classes of support data of T correspond bijectively to those of L .

Example: tt-spectrum

T : tt-category,

$L = \{\text{radicals of finitely generated tt-ideals}\}.$

Example 18

X_L is embedded injectively into $\text{Spec}_{\otimes}(T).$

$$\text{Ind}(L) = \{\text{radical tt-ideals}\},$$

$$X_L = \{P \in \text{Ind}(L) \mid \bigcap_{S \supsetneq P} S \not\supsetneq P\},$$

$$\text{Spec}_{\otimes}(T) = \{P \in \text{Ind}(L) \mid S \cap S' \not\supsetneq P \text{ for all } S, S' \supsetneq P\}.$$

Example: resolving subcategories

$R = S/(a_1, \dots, a_c)$: complete intersection ring,
 $V = \text{Sing}(\text{Proj}(S[x_1, \dots, x_c]/(a_1x_1 + \dots + a_cx_c)))$,
 $L = \{\text{finitely generated resolving subcategories of } D^b(R)\}.$

Example 19

$$\begin{aligned}\text{Ind}(L) &= \{\text{resolving subcategories of } D^b(R)\}, \\ X_L &= (\text{Spec } R \times \mathbb{N}) \times V.\end{aligned}$$

Results on the relative spectra

Localization

S : tt-category,
 T : S -module.

Proposition 1

Let $I \in \text{Sub}_S(T)$. Then

$$\text{Spec}(T/I, S) \cong \text{Spec}(T, S) \setminus \text{Supp}(I).$$

If I is finitely generated, then $\text{Spec}(T/I, S)$ is an open subset of $\text{Spec}(T, S)$.

$$\text{Sub}_S(T/I) \xleftarrow{1:1} \{J \in \text{Sub}_S(T) \mid J \supset I\}.$$

Comparisons

S : rigid tt-category.

Theorem 20

There is a natural dense embedding

$$\mathrm{Spec}(S, S) \rightarrow \mathrm{Spec}_{\otimes}(S),$$

where $\mathrm{Spec}_{\otimes}(S)$ is Balmer's tt-spectrum. If $\mathrm{Spec}_{\otimes}(S)$ is topologically noetherian, then this embedding is an isomorphism.

A prime submodule $P \subset S$ is a prime $\otimes$ -ideal.

In general, these spectra are not isomorphic. There is a counterexample.

Comparisons 2

T : S -module.

Theorem 21

If S is generated by 1_S as its own thick subcategory, then there is an isomorphism

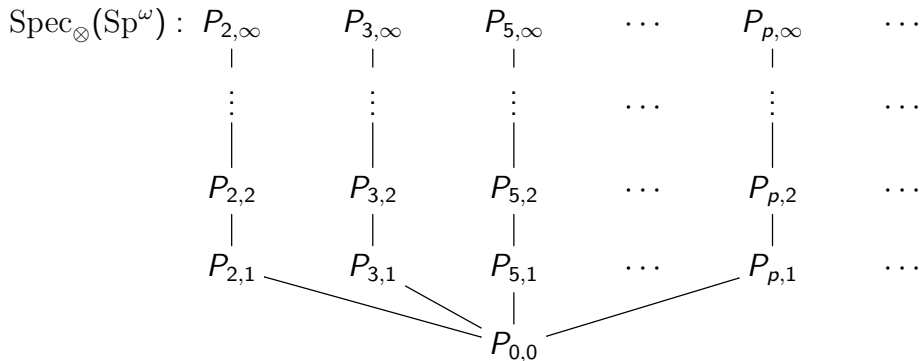
$$\mathrm{Spec}(T, S) \cong \mathrm{Spec}_{\Delta}(T),$$

where $\mathrm{Spec}_{\Delta}(T)$ is Matsui's triangular spectrum.

Example: category of finite spectra

Sp^ω : homotopy category of finite spectra.

Balmer 2010:



Example: category of finite spectra

Example 22

$$\begin{aligned}\mathrm{Spec}_{\Delta}(\mathrm{Sp}^{\omega}) &= \mathrm{Spec}(\mathrm{Sp}^{\omega}, \mathrm{Sp}^{\omega}) \\ &= \mathrm{Spec}_{\otimes}(\mathrm{Sp}^{\omega}) \setminus \{P_{2,\infty}, P_{3,\infty}, \dots\}.\end{aligned}$$

- 1 Sp^{ω} is classically generated by the unit object $\mathbb{S}$, hence Matsui's triangular spectrum and our space coincide.
- 2 Sp^{ω} is a rigid tt-category, hence our space embeds into Balmer's tt-spectrum, but they are not isomorphic.

Composability

D : stable ∞ -category.

Definition 23

Let $I, J \subset D$ be thick subcategories. We say that a pair (I, J) is *composable* if, for any $x \in I$, $y \in J$, any morphism $y \rightarrow x$ factors through some $z \in I \cap J$.

Let L_I be the Bousfield localization $\text{Ind}(D) \rightarrow \text{Ind}(D)$ with kernel $\text{Ind}(I)$. The pair (I, J) is composable if and only if

- 1 $L_I \circ L_J = L_{\langle I, J \rangle}$, and,
- 2 $\text{Ind}(I) \cap \text{Ind}(J) = \text{Ind}(I \cap J)$.

Composability

Both (I, J) and (J, I) are composable if and only if

- ① $L_I \circ L_J = L_J \circ L_I$, and,
- ② $\text{Ind}(I) \cap \text{Ind}(J) = \text{Ind}(I \cap J)$.

It is also equivalent to the following diagram being Cartesian:

$$\begin{array}{ccc}
 D/(I \cap J) & \longrightarrow & D/J \\
 \downarrow & & \downarrow \\
 D/I & \longrightarrow & D/\langle I, J \rangle.
 \end{array}$$

Thus, composability can be regarded as a "half" of commutativity.

Admissible module

S : tt- ∞ -category,

D : S -module (stable ∞ -category).

For a specialization-closed subset $Z \subset \operatorname{Spec}(S, S)$, set

$$S_Z := \{s \in S \mid \operatorname{Supp}(s) \subset Z\}, \quad D(Z) := \langle S_Z \otimes D \rangle.$$

Definition 24

A S -module D is called *admissible* if

- 1 $\operatorname{Spec}(S, S)$ is topologically noetherian.
- 2 For any closed subset $Z \subset \operatorname{Spec}(S, S)$ and any S -submodule $I \subset D$, the pair $(I, D(Z))$ is composable.
- 3 $D(Z \cap W) = D(Z) \cap D(W)$ for any closed subsets $Z, W \subset \operatorname{Spec}(S, S)$.
- 4 $D(\bigcup_i Z_i) = \langle \bigcup_i D(Z_i) \rangle$ for any family of closed subsets $\{Z_i\}_i \subset \operatorname{Spec}(S, S)$.

Admissible module

Definition 25

Assume $\mathrm{Spec}(S, S)$ is topologically noetherian and S is rigid.

- ① We say that D is *rigid* if there exists a natural equivalence

$$\mathrm{Hom}^0(s \otimes d, d') \cong \mathrm{Hom}^0(d, s^\vee \otimes d')$$

for all $s \in S$ and $d, d' \in D$, where $s^\vee$ is the dual of s .

- ② We say that D is *smashing* if, for any closed subset $Z \subset \mathrm{Spec}(S, S)$ and any $F \in D$, the natural morphism

$$L_{S_Z}(1_S) \otimes F \rightarrow L_{D(Z)}(F)$$

is an isomorphism.

Proposition 2

rigid $\Rightarrow$ smashing $\Rightarrow$ admissible.

Admissible module

D : admissible S -module.

Lemma 26

For $P \in \operatorname{Spec}(D, S)$, the set $\{s \in S \mid s \otimes D \subset P\}$ is a prime submodule of S .

Let $\operatorname{Spec}^{\text{fin}}(D, S)$ be the topological space whose closed subsets are generated by

$$\{\operatorname{Supp}(s \otimes D) \mid s \in S\} \cup \{\operatorname{Supp}(d) \mid d \in D\}.$$

If D is classically finitely generated, then $\operatorname{Spec}^{\text{fin}}(D, S) \cong \operatorname{Spec}(D, S)$, but in general the topology on $\operatorname{Spec}^{\text{fin}}(D, S)$ is finer than that on $\operatorname{Spec}(D, S)$.

Theorem 27

There is a natural morphism $\operatorname{Spec}^{\text{fin}}(D, S) \rightarrow \operatorname{Spec}(S, S)$.

Corollary 28

If S is rigid, there is a natural morphism $\operatorname{Spec}^{\text{fin}}(D, S) \rightarrow \operatorname{Spec}_{\otimes}(S)$.

Admissible module

D : admissible S -module.

For specialization-closed subsets $Z \subset W \subset \operatorname{Spec}(S, S)$, put
 $D_{W \setminus Z} := D(W)/D(Z)$.

Theorem 29

$$\operatorname{Spec}(S, S) \supset U \longmapsto \operatorname{Sub}_S(D_U) \in \mathbf{Set}$$

is a sheaf on $\operatorname{Spec}(S, S)$.

Theorem 30

There is a natural continuous bijection

$$\operatorname{Spec}(D, S)_{W \setminus Z} \rightarrow \operatorname{Spec}(D_{W \setminus Z}, S),$$

where $\operatorname{Spec}(D, S)_{W \setminus Z}$ is the topological fiber over $W \setminus Z \subset \operatorname{Spec}(S, S)$.

Example: Severi-Brauer scheme of relative dimension 1

Y : scheme,

X : Severi-Brauer scheme of relative dimension 1 over Y .

$$\mathrm{Spec}(\mathrm{Perf}(X), \mathrm{Perf}(Y)) \cong X \coprod \left(\coprod_{\mathbb{Z}} Y \right)$$

Example 31

The above is calculated as follows.

(Step 1) If Y is a spectrum of an algebraically closed field, it is due to Matsui.

(Step 2) We can show that the right-hand side is a subspace of the left-hand side.

(Step 3) To prove the both side are equal, we use fiberwise decomposition. Let $y \in Y$. We may show that every point of $\mathrm{Spec}(\mathrm{Perf}(X), \mathrm{Perf}(Y))_y$ "comes from" $\mathrm{Spec}(\mathrm{Perf}(X_{\bar{y}}), \mathrm{Perf}(\mathrm{Spec} \kappa(\bar{y}))) \cong X_{\bar{y}} \coprod (\coprod_{\mathbb{Z}} \mathrm{Spec} \kappa(\bar{y}))$.

Admissible module

S' : tt- ∞ -category, S : S' -linear tt- ∞ -category,
 D : admissible S -module.

Theorem 32

*Assume that S is $\mathrm{Spec}(S, S)$ -locally classically generated by the image of S' .
Then there is an embedding*

$$\mathrm{Spec}(D, S) \rightarrow \mathrm{Spec}(D, S').$$

If S itself is classically generated by the image of S' , then this morphism is an isomorphism. If D is a finitely generated S' -module, then this morphism is an open embedding.

Example: derived category of perfect complexes

$f : X \rightarrow Y$: morphism of schemes.

Example 33

The action of $\mathrm{Perf}(Y)$ on $\mathrm{Perf}(X)$ is rigid. There is a natural open immersion

$$\iota_X : X \rightarrow \mathrm{Spec}(\mathrm{Perf}(X), \mathrm{Perf}(Y))$$

and f decomposes as $\pi \circ \iota_X$, where

$\pi : \mathrm{Spec}(\mathrm{Perf}(X), \mathrm{Perf}(Y)) \rightarrow \mathrm{Spec}_{\otimes}(\mathrm{Perf}(Y)) \cong Y$ is the canonical morphism.

The space $\mathrm{Spec}(\mathrm{Perf}(X), \mathrm{Perf}(Y))$ contains all relative Fourier–Mukai partners of X over Y as open subschemes.

Example: derived category of perfect complexes

$f : X \rightarrow Y$, $g : Y \rightarrow Z$: morphisms of schemes.

Example 34

The following diagram is commutative:

$$\begin{array}{ccccccc}
 X & \xrightarrow{\iota_X} & \mathrm{Spec}(\mathrm{Perf}(X), \mathrm{Perf}(Y)) & \xrightarrow{s} & \mathrm{Spec}(\mathrm{Perf}(X), \mathrm{Perf}(Z)) & \xrightarrow{t} & \mathrm{Spec}_\Delta(\mathrm{Perf}(X)) \\
 & \searrow f & \downarrow & & \downarrow & & \downarrow \\
 & & Y & \xrightarrow{g} & Z & \xrightarrow{\mathrm{can.}} & \mathrm{Spec} \mathbb{Z}.
 \end{array}$$

The upper horizontal arrows are open immersions. If f (resp. g , resp. Z) is quasi-affine, then ι_X (resp. s , resp. t) is an isomorphism.

$$\text{tt-spectrum} \xleftarrow{\text{relative triangular spectrum}} \text{triangular spectrum}$$

Structure sheaf

Structure sheaf

k : tt- ∞ -category,

S : k -linear tt- ∞ -category,

D : S -module.

Let $\mathcal{O}_{(D,S,k)}$ be the sheaf on $\mathrm{Spec}(D, S)$ associated to

$$U \longmapsto \mathrm{Nat}_k^0(\mathrm{Id}_{D_U}, \mathrm{Id}_{D_U}),$$

where $D_U = D / \{a \in D \mid \mathrm{Supp}(a) \cap U = \emptyset\}$. We define a ringed space by

$$\mathrm{Spec}(D, S, k) := (\mathrm{Spec}(D, S), \mathcal{O}_{(D,S,k)}).$$

Structure sheaf

Some of the previous results can be extended to the setting of ringed spaces.

Example 35

- ① Let $f : X \rightarrow Y$ be a morphism of schemes. Then the following diagram of ringed spaces is commutative:

$$\begin{array}{ccc} X & \xrightarrow{\iota_X} & \mathrm{Spec}(\mathrm{Perf}(X), \mathrm{Perf}(Y), \mathrm{Perf}(\mathbb{Z})) \\ & \searrow f & \downarrow \pi \\ & & Y \end{array}$$

- ② Let S be a rigid tt- ∞ -category such that $\mathrm{Spec}_{\otimes}(S)$ is topologically noetherian. Then

$$\mathrm{Spec}_{\otimes}(\mathrm{Ho}(S)) \cong \mathrm{Spec}(S, S, S).$$

Thank you!